

Checkpoint 6: Parameter-Matched Capacity Control

Objective

Check whether the main architecture comparison is explained by parameter count.

Key Message

The capacity-control result makes the final claim honest: residual is not best on every metric, but remains best on SSIM, Delta E, and entropy.

Completed Work

- Completed parameter-matched U-Net capacity control with base_channels=42.
- Compared parameter-matched U-Net with the residual model.
- Qualified the final conclusion using metric-specific capacity-control evidence.

Evidence

Metric	Matched U-Net	Residual	Better
MSE	0.034294	0.035727	Matched U-Net
PSNR	16.356273	16.304393	Matched U-Net
SSIM	0.746291	0.788963	Residual
Delta E	27.225774	26.840257	Residual
Entropy	4.790907	4.368050	Residual

Notes

- The residual model leads on SSIM and Delta E; the matched U-Net leads on MSE, MAE, and PSNR.



Matching parameter-matched U-Net and residual restoration examples.